



Inspiring Excellence

BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: *Pet Health monitoring and automated feeding system*

by

Group_No 4
Section 8

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Abstract

Introducing the IoT enabled Pet Health Monitoring and Automated Feeding System that will cater to the health of pets for the pet owners. This system is based on Arduino Uno that is used for automated food dispensing systems, environment monitoring and tracking pet movements. In this work, we discussed the automatic feeding process which is controlled by Arduino to dispense food to the pet at specific time, DHT11 sensor measures surrounding temperature and humidity and activates a DC fan if the surrounding temperature and humidity are hazardous, and the use of infrared sensors to detect the movement of the pet and also send an alert if the pet is unable to use the food dispenser for 24 hours by the use of Bluetooth module and buzzer. The results produced confirm the solution concept as a viable and viable solution.

Keywords

microcontroller, automation, IoT, pet-health, embedded systems

1. Introduction

It's challenging to keep a constant eye on pet's eating and living habits in the modern busy lifestyle. Lack of monitoring for extended periods can result in feeding times being confused, unrecorded illnesses and risky interactions with the environment. The objective of the project is to design a smart and automated system for telemonitoring and remote maintenance of domestic pets like birds, cats and dogs.

The project will involve automatic feeding, real time environment control (humidity and temperature), monitoring pet activity around feeding station, and sending real time information to the owner's smart phone through bluetooth technology. The project is very much associated with Computer Interfacing as it involves the combination of a microcontroller (Arduino Uno) with different analog and digital sensors, actuators and communication protocols to develop a responsive embedded system..

2. Related Work/ Inspiration

Most of the classic automatic feeders use simple mechanical timers that control food distribution, but cannot monitor health or environmental conditions. There are premium commercial smart feeders that can be integrated with app and have WiFi cameras, but they are really expensive and don't control the immediate surroundings of the pets.

The concept for the project came when everyone realized that an integrated and low-cost solution is needed. Combining environmental control (24 hours auto fan on/off for temperature control), and safety check algorithms (24 hours no food intake alert system with IR sensor, 24 hours no water intake alert system with Water level sensor), it is one of a kind. This multi-dimensional approach not only ensures the pet is fed, but actively helps to keep the pet comfortable and notifies the pet owner to look for a health anomaly that may

have appeared (such as sudden loss of appetite).As well as helping with hygiene, by stopping the dispenser to make sure the food bowl is full so it overflows.

3. Technical Approach

System Architecture:

The Arduino Uno is used as the main processor.

- **Inputs:**DHT11 (environment), IR sensors (food intake and level tracking) and Water level sensor(water tracking
- **Outputs:** Servo motor (dispenser lid), DC Fan (cooling), LCD (Local display), Buzzer (Local alarm), Bluetooth Module (Remote alerts)

Components Used:

- **Arduino Uno:** The core microcontroller processing sensor data and controlling actuators.
- **Sensors:** IR Sensors (detects pet presence), DHT11 (measures temperature/humidity), Water Level Sensor.
- **Actuators & Outputs:** Servo Motors (rotates 90° to open/close food containers), DC Fan (cools the environment), LCD Monitor (displays system status locally), Buzzer (audible alerts).
- **Communication:** Bluetooth Module (transmits data to the owner's smartphone).

Cost Breakdown:

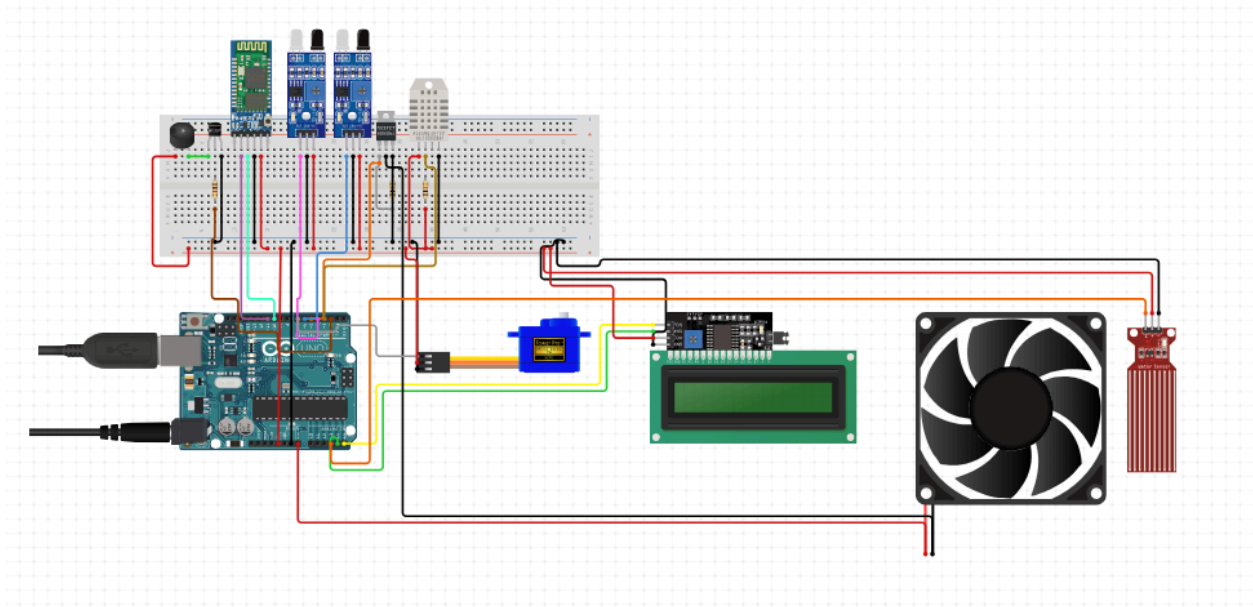
Component	Unit Cost (BDT)	Quantity	Total (BDT)
Arduino Uno	690	1	690
IR sensor	45	2	90
Servo motor	143	2	286
DC fan	110	1	110

LCD Monitor	195	1	195
Bluetooth module	349	1	349
Temp & Humidity (DHT11)	120	1	120
Buzzer	15	1	15
Breadboard	90	1	90
5V Power Adapter	160	1	160
Water level sensor	49	1	49
Total Approximate Cost			2,154

Communication Protocols:

1. **UART/Serial Communication:** Utilized by the Bluetooth module to transmit alerts to the smartphone app and used during the debugging phase via the Serial Monitor.
2. **PWM (Pulse Width Modulation):** Deployed to accurately control the servo motor's angular position (rotating exactly 90° for food dispensing) and to regulate the speed of the DC fan.
3. **I2C (Inter-Integrated Circuit):** Used for the LCD monitor, significantly reducing the wiring complexity by requiring only the SDA and SCL lines.

Circuit/Schematic:[circuit/schematic](#)



Implementation Steps

Time Based Feeding: The Arduino has a timer continuously running. It pulsates the duty cycle PWM signal to its servo motor at predetermined intervals to release the dose for a specified amount of time, but doesn't release it past a maximum dose to avoid overfeeding. The DHT11 is an ambient temperature and humidity sensor. If thresholds are exceeded, the DC fan is activated (For now, we used the humidity threshold which is easier to control and better for demonstration where exceeding 75% humidity turns the fan on) the bluetooth and LCD will show the humidity and temperature live. Well, Fan status is also shown (Serving as a warning that the threshold is exceeded).

A timer is always running in the background (Inactivity Alert). When the pet is detected by the IR sensor, the timer restarts. After 24 hours, without an IR trigger, the buzzer will beep, and an alert notification will be sent to the owner via Bluetooth.

Challenges:

Controlling the power consumption was one major issue – the servo, DC fan and Bluetooth module could not be powered at the same time without losses of voltage. The solution was to make sure that the 5V power adapter was capable of supplying enough current, and by incorporating decoupling capacitors on the breadboard. Another, was finding a reliable supplier. Many of the components we bought were first faulty (bluetooth module, DHT11, servo etc.). So we had to purchase them again in the end. Which has given us a lot of delay and problems.

4. Sustainability & Impact

Sustainability: : Exactly the correct amount of food is discharged so as to minimise food wastage and food spoilage. In addition, the system uses low powered sensors combined with a microcontroller that makes sure that high power consuming devices, such as DC fans, are used selectively and only when needed. The casing of the device can be recycled..

Impact: The system is very helpful for pet owners who work, and it helps to ease their worries and concerns about their cat/dog's health. The mechanism helps detect changes in behavior such as not eating for over a day and prevent serious diseases in pets.

Future Work: WiFi technologies could be added for global control, camera modules for visual confirmation and solar panels for energy independence in the next generation of the system

Limitations: The current version works only with Bluetooth technology, which has a limited range (up to 10 meters) and thus the pet owners might not be able to receive verified notifications while being far from home. The system would not work if the power supply is interrupted.

5. Results & Discussion

All major objectives of the prototype were achieved. The servo motor always turned 90° at the correct moments and would never jam food being dispensed. When a simulated humidity was added, and the environment was changed, the DHT11 sensor accurately reflected the changes, and the DC fan was immediately turned on and worked successfully to reduce the local humidity and temperature. Furthermore, the 24hr IR absence logic and 24hr water level not decrease logic were both perfect during testing on a shorter time scale (e.g., 1 minute intervals for debugging) and resulted in triggering of both the buzzer and the smartphone bluetooth alert. The LCD display perfectly matches the serial monitor display.

Below is a table that illustrates how the high prices decrease the likelihood of adoption of pets or the use of automated pet feeders.

Market Restraint	Impact Rank	Negative CAGR Impact	2026-2028	2029-2031	2032-2034
High product costs and affordability concerns	High	-1.6%	High	Medium	Low
Data privacy and connectivity concerns	Medium	-1.2%	Medium	High	Medium
Limited consumer awareness in emerging markets	Low	-0.9%	Medium	Low	Low
Others	Lowest	-0.5%	Low	Low	Medium
Total Restraint Impact		-4.2%			

6. Conclusion

The Pet Health Monitoring and Automated Feeding System is used to successfully demonstrate the use of microcontroller interfacing with a real-world problem. The project has integrated the time scheduled feeding, DHT11 environmental control, and IR food volume monitoring functions, and has comprehensively developed a smart pet care system with a reasonable cost. The design's technical prowess is demonstrated by the successful implementation of protocols like UART, PWM, and I2C. In conclusion, this project offers a successful demonstration that it is feasible to create a sustainable and automatic pet care system.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Md. Rafi Uddin Auntik	22201075	Hardware design, Circuit building, code, etc	- Contributed in the C code writing for various components - Contributed in the circuit building - Primary circuit design and component list making

			● Report writing
Naved Abrar Nibir	22201025	Software Design,Component buying, code,etc	● Contributed in the C code writing for various components ● Contributed in the circuit building ● Bought the components for the project. ● Report writing
Md Saidul Islam Apu	21301668	Hardware Design,Component buying, code,etc	● Contributed in the C code writing for various components ● Contributed in the circuit building ● Report writing
Fariha Yasmin	22301331	Software Design,Hardware design,code,etc	● Contributed in the C code writing for various components ● Contributed in the circuit building ● Report writing

References

- *VERY EASY Arduino IR Sensor Tutorial for Beginners | IR Sensor Arduino Tutorial | Infrared Sensor.* (n.d.). WwW.Youtube.Com. Retrieved August 30, 2026, from <https://www.youtube.com/watch?v=OMZacCLRt9A>
- Fact. MR. (2024, August 20). *Automatic pet feeder market share and statistics – 2034* . Fact. MR . <https://www.factmr.com/report/automatic-pet-feeder-market>
- Alvin Pribadi . (2026) . *Arduino DHT11 Temperature & Humidity Sensor with I2C LCD | Complete Tutorial (Free Code)* [Video]. YouTube. <https://www.youtube.com/watch?v=LeaTcP8VddI>
- Food52 . (2025, September 8) . *Automatic feeders: The benefits & considerations for your pet.* Food52 . <https://food52.com/story/pros-and-cons-of-automatic-feeders>
- *WATER LEVEL SENSOR with Arduino UNO | How to use WATER LEVEL SENSOR ARDUINO [Code & Circuit Diagram].* (n.d.). WwW.Youtube.Com. Retrieved August 30, 2026, from <https://www.youtube.com/watch?v=GnD-hRnWFLA>
- Market Growth Reports. (2026, July 29). *Smart pet products market size, trend & growth [2025 to 2033]*. Market Growth Reports. <https://www.marketgrowthreports.com/market-reports/smart-pet-products-100201>
- *How to Control a Servo With an Arduino.* (n.d.). WwW.Youtube.Com. Retrieved August 30, 2026, from <https://www.youtube.com/watch?v=QbgTl6VSA9Y>

